

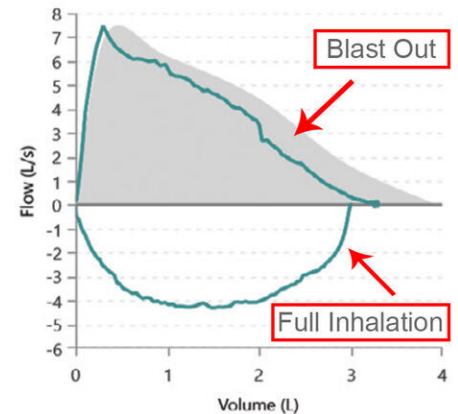
FVC Forced Vital Capacity

Setup:

Ensure HHU is detached from MiniBox+ and allow the HHU to calibrate. Hand the Patient the HHU and have the patient seal their mouth around the mouthpiece (teeth over the mouthpiece • tight seal with lips) & apply nose clip to the patient.

Coaching:

- Start with 2-3 normal relaxed breaths
- Deep inhale and immediate forced fast exhale
- Squeeze all air out until wheel turns **GREEN**
- Deep breath back in (close flow volume loop)



Example: Flow Volume Loop

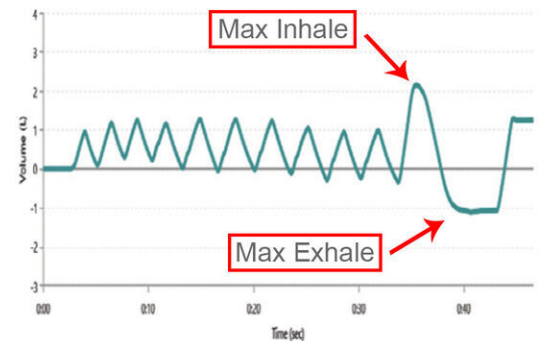
LVM Lung Volume Measurement

Setup:

Ensure HHU is attached to MiniBox+ & adjust the height of the device adequately for the patient. Have the patient seal their mouth around the mouthpiece (teeth over the mouthpiece • tight seal with lips) & apply nose clip to the patient.

Coaching:

- Relaxed normal breathing for about a minute
- Nice & slow tidal breaths, in & out (keeping breaths below 1 Liter)
- Once the counter turns **GREEN** & reads "6" instruct the patient on the next breath to take a slow deep breath in to vital capacity (full)
- Then breathe out slow, squeeze all the air out until the line hits plateau (empty) & then take a breath in & come off mouthpiece



Example: Tidal Breath LVM

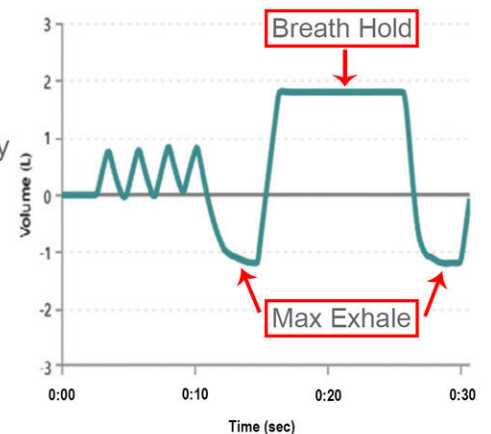
DLCO Diffusion Capacity

Setup:

Ensure HHU is attached to MiniBox+ & adjust the height of the device adequately for the patient. Have the patient seal their mouth around the mouthpiece (teeth over the mouthpiece • tight seal with lips) & apply nose clip to the patient.

Coaching:

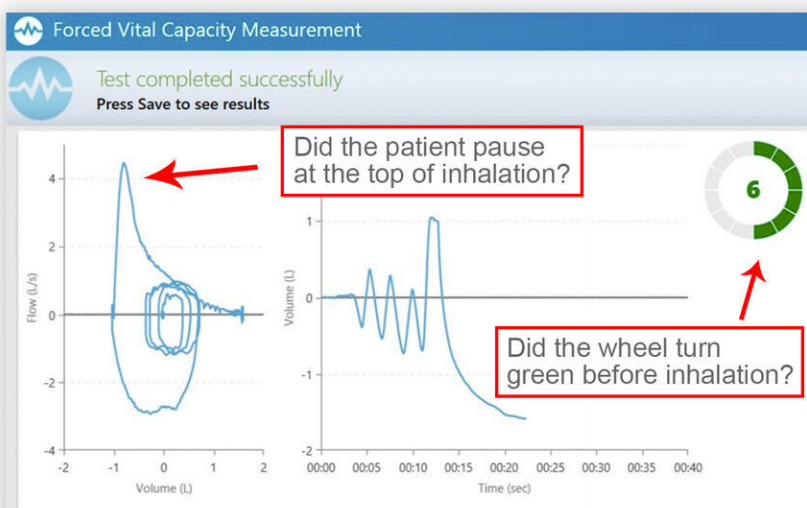
- Start with an inhale & take 4 relaxed tidal breaths & then exhale fully
- When the device clicks, take a fast deep breath in & HOLD when the inhalation is above the **GREEN** dotted line that appears
- Coach the patient to hold their breath for 8-10 seconds
- Once the wheel turns **GREEN** exhale completely to empty
- Once empty – inhale and come off mouthpiece to complete test



Example: Single Breath Diffusion

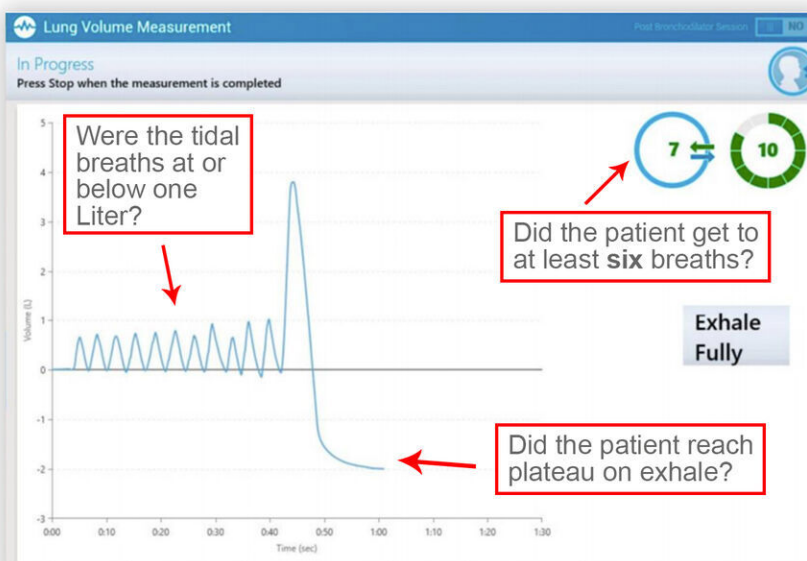
FVC Forced Vital Capacity

- ✓ Check: It is important that the inhalation be fast and that any pause during full inhalation be minimal.
- ✓ Check: The test assumes a full inhalation before beginning the forced exhalation and it is imperative that the patient takes a complete inhalation.
- ✓ Check: The patient should be prompted to “blast,” not “blow” the air from the lungs and should be encouraged to fully exhale.
- ✓ Check: It is important that the inhalation that follows the exhalation be as fast and as deep as possible.



LVM Lung Volume Measurement

- ✓ Check: It is important that the patient performs consistent relaxed normal breathing.
- ✓ Check: The LVM event counter to verify that **SIX** successful events were performed. After the counter has reached six events it turns green and the interruption valve deactivates automatically.
- ✓ Check: The patient should inhale deeply to their TLC level and then to exhale slowly and completely.
- ✓ Check: An acceptable maneuver is one where there is no evidence of leakage, which can be recognized by a downwards or upwards sloping graph.



DLCO Diffusion Capacity

- ✓ Check: The patient must start on an inhalation and complete FOUR normal breaths.
- ✓ Check: The patient exhales fully on the fourth breath.
- ✓ Check: A full inhale quickly to TLC and then hold their breath for 10 seconds.
- ✓ Check: Wait for the valve to close (device clicks) before instructing the patient to breathe in.
- ✓ Check: The patient is expected to inhale 85% of their vital capacity within 4 seconds.
- ✓ Check: Exhale completely and relatively rapidly to empty (wait 1 extra second before instructing them to breathe in).

